Avinabh Thakur

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INTERESTS

Electronics and electrical systems, embedded systems and firmware development, VLSI, Simulation, microelectronics and semiconductor devices, digital and analog hardware design, computer organization and architecture, hardware–software co-designs, Signal Processing Algorithms, RF Hardware Design, Robotics and control systems, applied programming for hardware, Artificial intelligence, embodied AI, reinforcement learning.

EMPLOYMENT HISTORY

Research Assistant -Engineer 2025-Present
HATRIC Labs, Nepal Technology Innovation Center, Kathmandu University

Defined electrical and safety requirements for a TENS system, including stimulation frequency range, current limits, and isolation considerations for biological tissue.

Built an H-bridge output stage using MOSFET which were switched using NPN BJTs to generate asymmetric biphasic stimulation by enabling controlled polarity reversal across the electrodes.

Integrated a passive RC low-pass filter to smooth the PWM waveform, reduce switching noise, and obtain a stable low-frequency stimulation signal.

Tested and validated the system using oscilloscope and multimeter measurements, analyzing waveforms at each stage and tuning component values for safe and reliable operation.

Documented complete circuit design, component selection, waveforms, and testing results for the TENS system in a structured technical report.

Contributed to patent documentation by preparing technical descriptions, block diagrams, and other support material related to the TENS device design.

SKILLS

Deep expertise in analog & digital electronics, CMOS circuits, signals & systems, EM fundamentals, noise, timing, power, and signal integrity,

Computer Organization and architecture: instruction sets, pipelining, caches, memory hierarchy, interrupts, DMA, I/O subsystems, performance optimization, Control and System design.

VLSI & semiconductor engineering: RTL-to-GDS flow, Verilog/SystemVerilog, timing & power analysis, physical design fundamentals, SPICE simulation, device physics, and IC fabrication process

Analog & mixed-signal design, op-amp architectures, ADC/DAC design, PLLs, clocking systems, stability, mismatch, and noise analysis

Sensors & instrumentation: design, modeling, interfacing, calibration, signal conditioning, and prototyping of custom and novel sensors; MEMS

RF, microwave & antenna engineering: transmission lines, impedance matching, S-parameters, RF front-end basics, antenna design, simulation, and measurement

Programmable & computing platforms: microcontrollers, processors, SoCs, FPGA/CPLD; hardware–software co-design, DSP pipelines, and acceleration concepts

Memory & architecture: SRAM, DRAM, Flash, ROM/EEPROM architectures, caching, latency, bandwidth, and power optimization

Embedded systems & low-level computing: C/C++, Python, ROS, Linux, RTOS concepts, DMA, interrupts, bus protocols, system debugging and verification, Microcontrollers

Proficient in RISC architecture principles, including instruction pipelining, load/store models, and single-cycle execution to optimize hardware efficiency and performance.

IoT & communication systems: sensor-to-cloud architectures, edge computing, wired and wireless communication fundamentals, protocol integration, and system-level optimization

Simulation & EDA: MATLAB/Simulink, Python (NumPy/SciPy), SPICE, Kicad, Proetus Cadence/Synopsys toolchains, EM and system-level simulation workflows

Embodied AI & robotics: reinforcement learning, perception–control pipelines, sensor fusion, real-time decision-making, and hardware-aware deployment on embedded and edge platforms

Research & innovation: IEEE literature analysis, experimental design, data-driven modeling, reproducible research, technical documentation, and engineering-led innovation.

Systems & decision engineering: engineering economics and valuation (cost modeling, ROI, NPV/IRR, lifecycle analysis, design trade-offs) supporting research-to-product decisions

EDUCATION

Bachelor of Engineering ~2027
Electrical and Electronics Engineering
Kathmandu University, Dhulikhel, Nepal
Computer Programming, Engineering Drawing, Advanced Physics, Chemistry, Engineering Management, Engineering Economics Digital Logic, Advanced Calculus, Diff. Equations, Probability & Stat, Electronics I & II, Network Analysis, Signals & Systems, Electrical Machinery, Microprocessors, Electromagnetics, Communication Systems, Measurement & Instrumentation, DSP, Biomedical Engineering,

High Schooling 2019- 2021
Science
Capital College and Research Center, Koteshwor, Kathmandu

CLUBS AND ORGANIZATIONS

Institute of Electrical and Electronics Engineering (IEEE)-KU Chapter 2026-present
IEEE Electron Device Society - KU
Chairman - Undergraduate

- Conducted Research paper writing workshops.
- Launched Magazine
- Organized events to connect research with industry.

Engineers Without Borders-KU chapter 2025-Present
Executive Member

- Designed website
- Partnered in Projects

Team UDAAN- Sounding Rocket 2024-2025
Avionics Researcher

- Designed on board Data acquisition system.
- Real time Communication to ground station using LoRA
- Signal conditioning and arrangement.

Society of Electrical and Electronics Engineers (SEEE) 2022-Present
Documentation Lead

- Organized workshops on using Engineering tools with 58 paid participants.
- Documented Events

Kathmandu University Robotics Club (KURC) 2022-Present
General Member

- Volunteered for the Aquabot event (Aavishkar 22).

Computer Science Initiatives in Nepal (CSIN) 2020-2022
Mentor

- Mentored members.
- Delivered a keynote talk on designing technology for older adults, reaching an audience of 200+ listeners.

CCRC Tech/IT Club 2019-2021
Vice President.

- Hosted science exhibitions and participated in the LOCUS event at IOE.

PROFESSIONAL ACTIVITIES

Training and Mentorship

- Trained high school students on basic electronic components, their symbols, functions, and real-world applications.
- Conducted interactive Arduino programming workshops for high school students also introducing fundamentals of robotics and automation.
- Conducted Documentation, Presentation and Pitching training to juniors and fellow university students.
- Conducted training and workshops on KiCad and Porteous simulation with 124 participants.
- Taught Mathematics to Grade 8 and 9 students as a tutor.
- Mentored Junior fellows in projects and Documentation.

Services

- Assisted in Robotics Club events, supporting workshops and competitions
- Designed circuit schematics using KiCad for departmental projects.
- Volunteered at the KU CAP Conference, helping with coordination and logistics.
- Reviewed and provided feedback on project documentation of junior students.
- Documented SEEE Events.
- Reviewer for multiple Documentations and Journals.
- Organized SEEE Hackathons, managing events and mentoring participants.
- Designed, Coded, Optimized and Launched EWB Website

AWARDS AND ACHIVEMENTS

Best Project for EEPX exhibition	1st Position	2024
Aavishkar'23 Hackathon "*Innovation Nexus*"	2nd Runner up	2023
National Informatics Olympiad	Bronze Medal	2022
International Youth Mathematics Challenge (IYMC) Bangladesh	World Rank 7th	2022
State Level Cricket, Janakpur	Man of the Match	2020

PROJECTS

1. Designed a Logic IC Based Binary Calculator that performed binary arithmetic operations using combinational logic circuits. (1st year Project)
2. Fabricated Binary Logic gates, counters, flipflops and oscillators using basic electronics components. (Hobby project)
3. Studied and Designed 4 channel transmitter and receiver using RF signals. (2nd year project.)
4. Developed a Bionic Arm Using Flex Sensors, enabling prosthetic hand movements controlled by finger bending. (KURC hackathon)
5. Implemented an Air Canvas Using Mediapipe in Python, allowing users to draw in the air through real-time hand gesture recognition. (Hobby project)
6. Built an RC Glider Using Flysky Six Channel T&R, demonstrating control systems and aerodynamics in a remote-controlled aircraft. (Clubs Joint Project)
7. Created a Self-Driving Algorithm Using Computer Vision to detect lanes and obstacles, enabling autonomous navigation in a prototype vehicle and implementing into a GPS based Delivery System. (Group Project)
8. Constructed a TOD Based Energy Meter, measuring energy consumption with variable rates based on the time of day. (Power System Mini Project)
9. Designed and Implemented a Noise-Canceling Algorithm on FPGA, reducing unwanted signals and demonstrating hardware-level signal processing. (3rd year project)
10. Developed MediBot, an AI-driven robotic system for autonomous wound stitching that integrates multimodal sensor fusion, custom neural networks for path planning, and an IoT monitoring layer for real-time remote supervision, (Harvard Health Hackathon.)

BOOKS

Authoring "Between 0 and 1" explains how electronics are created from raw sand to silicon, and from basic components to fully functional electronics to modern control and thinking Machines.

PATENTS

Applied For National Patent through HATRIC Lab for TENS machine

ENTREPRENEURSIP

Co-founder 2021 - 2023

PrepOp, Kathmandu, Nepal

- Created a platform for teachers and students where they can buy and sell courses, especially designed for Nepal's education system and internet infrastructure.
- Pitched to Multiple Investors and Educational institution.
- Demonstrated to Teachers and Students in Events.

PUBLICATIONS, JOURNALS AND ARTICLES

3. *Low-Power, High-Speed Digital Computation: A Modified Multiplier Approach for Optimizing Arithmetic and Logic Operations.*
2. *AI-Driven Deforestation Monitoring: A Multi-Modal Deep Learning Approach using Satellite Imagery and Reinforcement Learning*
1. *Beyond Line-of-Sight: An Interconnected V2X Prototype for Multi-Agent Semantic Awareness and Advanced Collision Avoidance and FSD.*

COMMUNITY SERVICES	Signed petitions and supported initiatives against corruption and for Gen Z empowerment, promoting transparency, accountability, and youth engagement in local governance and social issues.	2025
	Initiated campaigns and community projects around public health, environmental cleanliness, waste management, or local social issues, involving grassroots participation	2023
	Hosted workshops for traditional Nepali arts like pottery, painting, wood carving, or folk music and dance, promoting cultural preservation and local creativity	2022
	Conducted organized clean-up campaigns for rivers, streams, and surrounding areas to reduce pollution, protect aquatic life, and raise environmental awareness.	2020
	Taught older adults how to use smartphones, messaging apps, and the internet to help them stay connected with family, access services, and navigate modern life.	2020